ViGoEmotions

Fine-grained emotion recognition
in Vietnamese social media

What changes when Neutral becomes a learned output?

Three models. Two label spaces. One shared 20-epoch protocol.

3
model families

20
epochs for every neural run

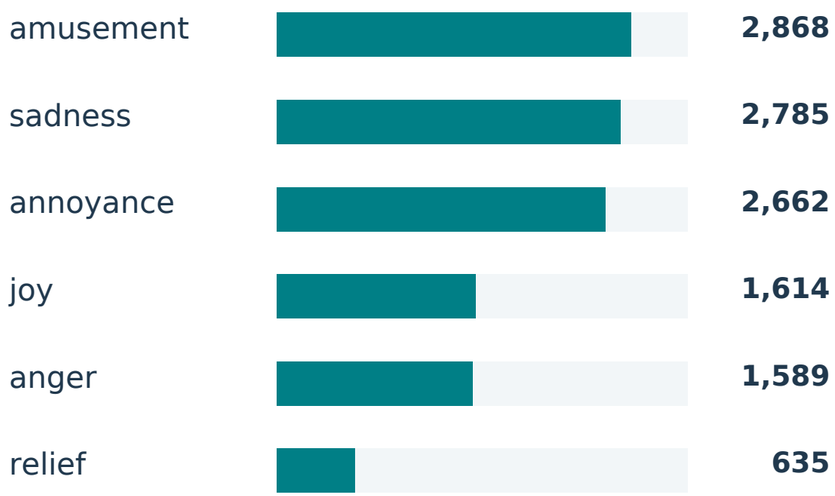
2
label settings: 27 and 28

Dataset, split and emotion frequency

20,664 source comments • fixed supplied partitions • label statistics from training data

Split	Source n	Used n	Role
Train	16,531	16,530	Fit model parameters
Validation	2,066	2,066	Select checkpoint / explore thresholds
Test	2,067	2,067	Held out from model evaluation
Total	20,664	20,663	One train–test text overlap removed

Most frequent five + least frequent label



1.859

emotions / train comment

11 / 47

median tokens / characters

TAKEAWAY

Preserve the split; fit text representations on train only.

Sources: current study annotation_counts.csv; fixed split IDs; PhoBERT-27 per-label table. No test model scores.

27 vs 28 outputs: one additional decision

The label mapping and validation examples stay fixed; Neutral becomes an explicit target.

Property	27-emotion model	28-label model
Output space	27 emotion decisions	27 emotions + Neutral
Neutral-only target	All 27 emotion entries are zero	Neutral = 1; emotion entries = 0
Mixed annotation	Retain the emotion entries	Retain emotions and Neutral
Reported full Macro-F1	Average over 27 labels	Average over 28 labels

SAME INPUT, DIFFERENT TARGET SPACE

Target: joy + Neutral → **27 outputs: joy = 1** | **28 outputs: joy = 1, neutral = 1**

TAKEAWAY **A missing emotion prediction is not automatically a Neutral prediction.**

Sources: config27.json; evidence/28/config.json; annotation_counts.csv. Encoding examples are schematic targets.

Three routes from text to multi-label scores

Model-specific text representations • 27 or 28 independent label decisions

TF-IDF + LR Lexical baseline	**Regex tokens** 1–2 grams	→	**TF-IDF** Fit vocabulary on train	→ **OvR classifiers** One LR per label → **Label scores** Threshold at 0.5
BIGRU Learn from scratch	**Token IDs** Lowercase + whitespace	→	**Embedding** 128 dimensions	→ **BiGRU + linear** Forward + backward → **Sigmoid scores** 27 / 28 outputs
PHOBERT Full fine-tuning	**Segment + BPE** VnCoreNLP tokenizer	→	**PhoBERT backbone** Pretrained Vietnamese	→ **Classification head** Train with backbone → **Sigmoid scores** 27 / 28 outputs

Shared preprocessing: Unicode NFC + whitespace normalization; retain Vietnamese accents and negation.

TAKEAWAY **Text representation changes across models; the label decision remains independent for each output.**

Sources: frozen run configurations; train_neural.py; neutral_28_labels.py. Neural training: BCEWithLogitsLoss.

Architectures and training configurations

Shared seed 42 • fixed split • threshold 0.5 • Tesla T4 for neural models

Model	Representation / architecture	Parameters 27 / 28	Batch / learning rate	Budget
TF-IDF + OvR LR	Unigram/bigram; up to 40k features; liblinear, C = 1	0.941M / 0.975M	Not neural	max_iter = 1,000
BiGRU	Train from scratch; embed/hidden 128; length 128	2.128M / 2.129M	64 / 0.001	20 epochs
PhoBERT	VnCoreNLP + BPE; full fine-tuning; length 256	135.019M / 135.020M	16 / 2e-5	20 epochs

Select by validation emotion27 Macro-F1 at 0.5, including for the 28-output model; ties keep the earliest epoch.

TAKEAWAY **Equal epochs control the training budget; they do not imply equal compute.**

Sources: frozen manifests and six run.json files. Parameter counts shown as 27 / 28 outputs; rounded to millions.

PhoBERT leads within each label setting

Validation n = 2,066 • threshold 0.5 • separate full-label evaluations for 27 and 28 outputs

Labels	Model	Macro-F1 ↑	Micro-F1 ↑	Hamming ↓	LRAP ↑	Best ep.	Train time
27	TF-IDF + OvR LR	0.2388	0.2712	0.0606	0.6407	—	2.75 s
27	BiGRU	0.4220	0.4590	0.0626	0.6191	17	80.3 s
27	**PhoBERT**	**0.5974**	**0.5996**	**0.0550**	**0.7419**	14	4.05 h
28	TF-IDF + OvR LR	0.2310	0.2662	0.0601	0.6096	—	2.52 s
28	BiGRU	0.4086	0.4483	0.0628	0.5851	20	83.6 s
28	**PhoBERT**	**0.5783**	**0.5829**	**0.0575**	**0.7077**	15	3.90 h

Both neural models completed 20 epochs. “Best ep.” identifies the selected checkpoint, not the number of epochs trained.

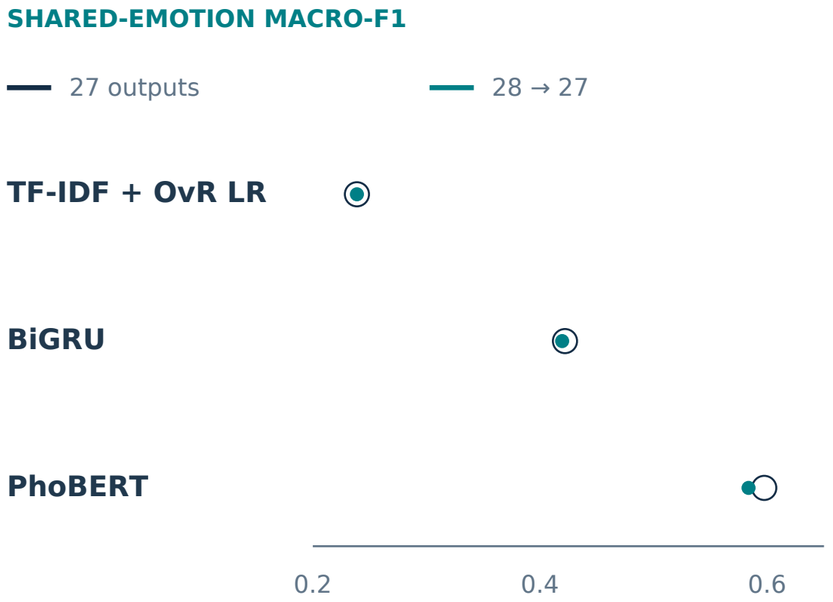
TAKEAWAY **Rank models within each block; the two Macro-F1 blocks evaluate different label sets.**

Source: model_comparison.csv. ↑ higher / ↓ lower is better. CPU TF-IDF; T4 neural. Times exclude preprocessing.

Adding Neutral does not raise shared-emotion F1 here

Same 27 emotion columns • same validation IDs • shared checkpoint selection • one seed

Model	27-output F1	28 → 27 F1	Δ (pp)
TF-IDF + OvR LR	0.2388	0.2388	+0.00
BiGRU	0.4220	0.4195	-0.25
PhoBERT	0.5974	0.5836	**-1.38**



This controls the epoch budget and scoring label space. It does not establish a general causal effect or statistical significance.

TAKEAWAY **PhoBERT changes by −1.38 percentage points on the shared 27-emotion evaluation.**

Source: paired_emotion27_delta.csv. Δ = 28-output projection minus 27-output score, in percentage points.

A learned Neutral output still makes substantial errors

28-label setting only • 94 Neutral-positive validation examples: 75 Neutral-only + 19 mixed

Model	Precision	Recall	F1	TP	FP	FN
TF-IDF + OvR LR	**0.5000**	0.0106	0.0208	1	1	93
BiGRU	0.1778	0.0851	0.1151	8	37	86
PhoBERT	0.4018	**0.4787**	**0.4369**	45	67	49

45
correct Neutral detections

49
missed Neutral examples

67
false Neutral predictions

An empty prediction is not a substitute for an explicitly trained Neutral label.

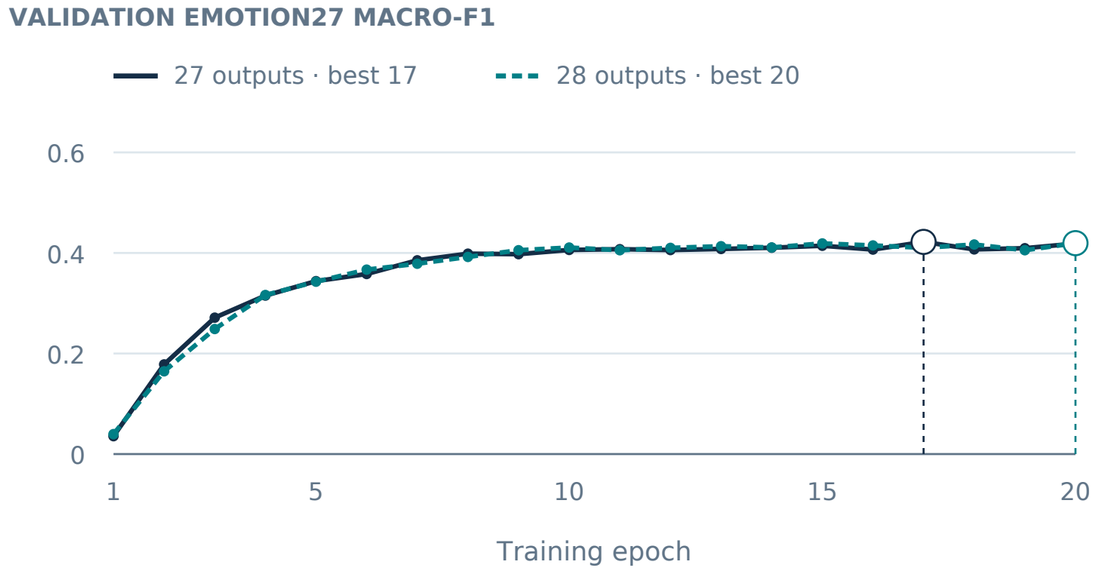
TAKEAWAY **PhoBERT finds 45 of 94 Neutral examples, with 67 false alarms.**

Source: neutral_metrics.csv. Threshold 0.5; support 94 for all models. No Neutral metric for the 27-output model.

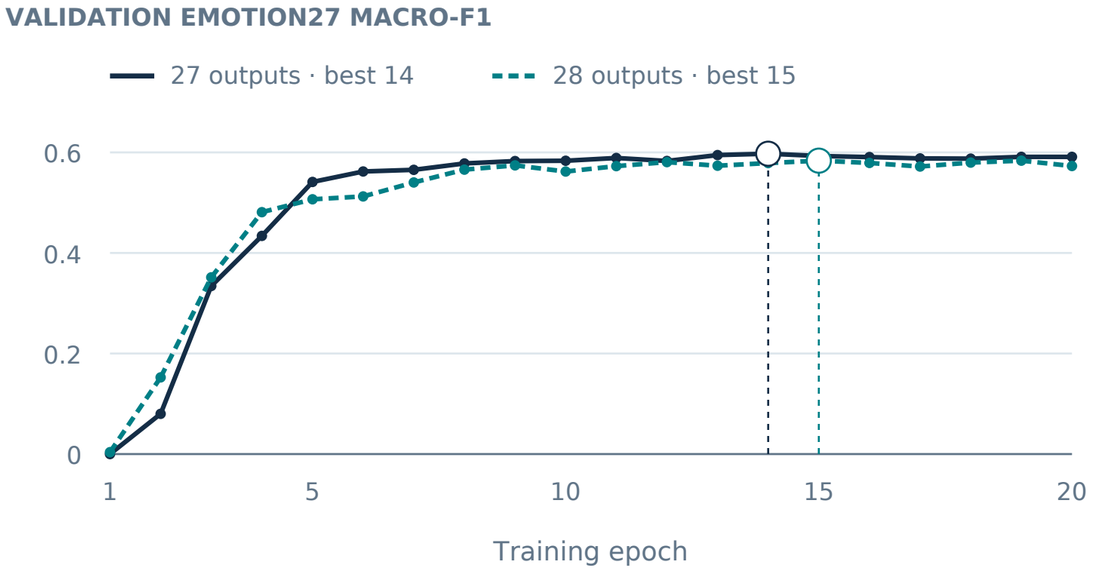
Twenty epochs trained; the last checkpoint need not win

Both curves in each panel use the same 27-emotion checkpoint-selection metric.

BiGRU



PhoBERT



Train BCE continued to decrease; validation loss was not recorded. These curves alone do not confirm overfitting.

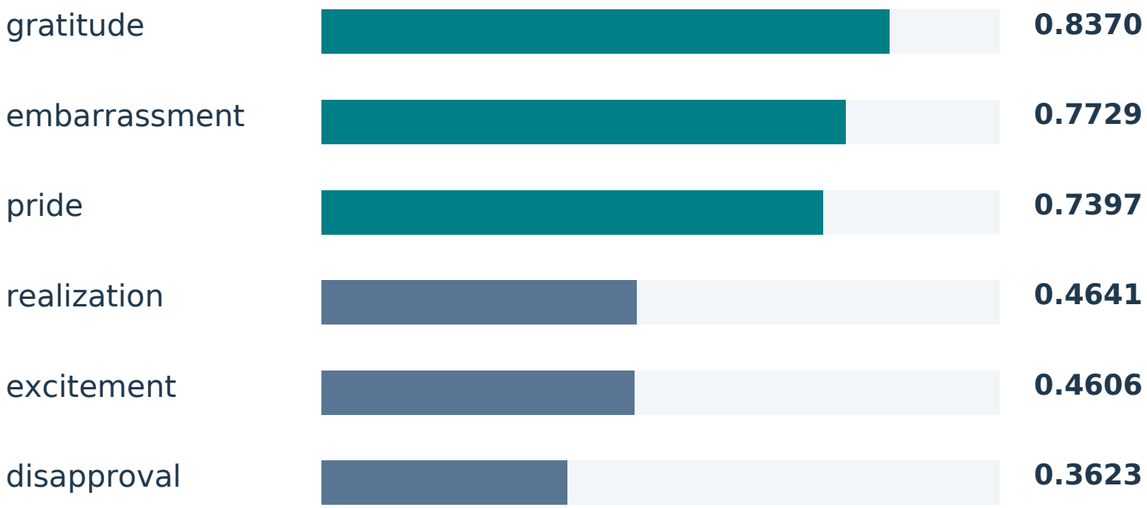
TAKEAWAY **Best PhoBERT checkpoints: epoch 14 for 27 outputs and epoch 15 for 28 outputs.**

Sources: evidence/{27,28}/{gru,phobert}/run.json. All histories contain epochs 1–20; no early stopping.

Performance varies by label, beyond raw frequency

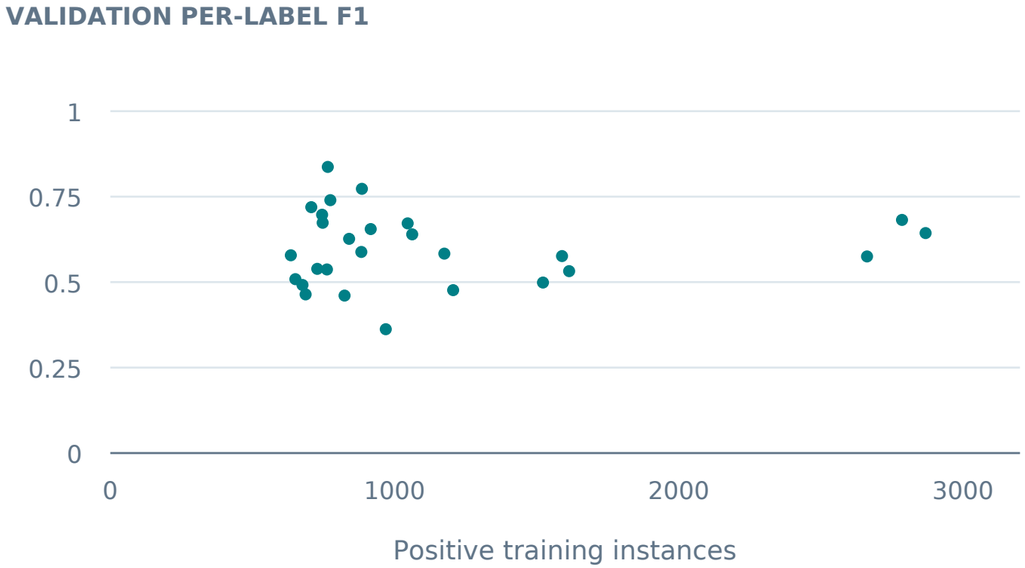
PhoBERT, 27 outputs • per-label validation F1 at threshold 0.5

Q1 / HIGHEST AND LOWEST PER-LABEL F1



Selected top three and bottom three labels; full 27-label results remain in the report.

Q2 / TRAINING FREQUENCY VS PER-LABEL F1



Spearman ρ = 0.018; p = 0.928

TAKEAWAY

Gratitude is strongest; disapproval is weakest. Frequency shows little monotonic association in this run.

Sources: per_label_metrics.csv; frequency_f1.csv; current PhoBERT-27 per-label table. Descriptive, single-seed analysis.

Research questions: a compact evidence checklist

Current 27-output study • frozen checkpoints • validation-only diagnostics

RQ	What was measured?	Selected result	Interpretation / boundary
Q1	Highest / lowest label F1	Gratitude 0.8370; disapproval 0.3623	PhoBERT-27; full label table in report
Q2	Training frequency vs F1	Spearman ρ 0.018; p 0.928	Descriptive association, not causation
Q3	Disappointment + sadness	PhoBERT: 63/122 jointly correct (51.64%)	Pair chosen from train: 904 occurrences
Q4	PhoBERT advantage by style	Teencode: 0.5547 vs GRU 0.3333; +22.13 pp	Largest gap among heuristic groups; n = 307
Q5	Fixed vs explored threshold	τ 0.50 → 0.35: Macro-F1 0.5974 → 0.5978	Tiny F1 gain; Hamming Loss worsens

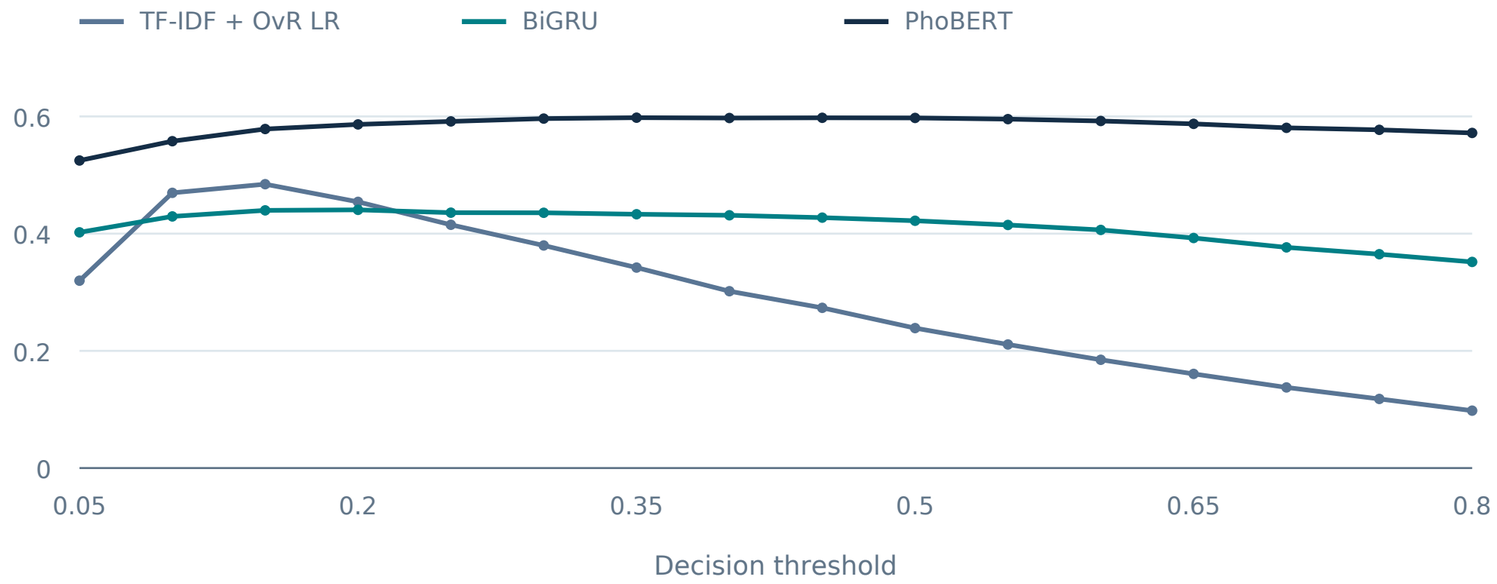
TAKEAWAY **The findings depend on the label, text style, metric and decision threshold.**

Sources: label_extremes, frequency_f1, pair_preservation, neural_style_comparison, threshold_sensitivity.csv.

PhoBERT gains little from this global threshold grid

27 outputs • same checkpoint • thresholds explored on validation only

VALIDATION MACRO-F1



0.5974

Macro-F1 at τ = 0.50

0.5978

Macro-F1 at τ = 0.35

+0.043 pp

exploratory F1 difference

Hamming Loss rises from 0.0550 to 0.0587; LRAP is independent of the cutoff.

TAKEAWAY **Keep the main comparison at 0.5; test generalization remains unknown.**

Source: current analysis/27_emotion27/*/tables/*_thresholds_val.csv. Grid 0.05–0.80, step 0.05.

Neutral errors: implied affect versus surface cues

Two illustrative cases from the six-case preliminary, AI-assisted review; no raw comments reproduced.

FALSE POSITIVE / NEUTRAL

Validation ID 30099

Annotated: relief

Predicted: Neutral

P(Neutral) = 0.9965

Hypothesis: implied emotion in an explanatory comment.

FALSE NEGATIVE / NEUTRAL

Validation ID 6238

Annotated: Neutral

Predicted: anger + annoyance

P(Neutral) = 0.00005

Hypothesis: profanity may resemble negative affect.

Review scope: three confident Neutral false positives and three low-probability Neutral misses, selected deliberately.

TAKEAWAY

Interpretations are hypotheses from selected cases, not established error causes.

Source: phobert28_error_review.csv; current PhoBERT-28 validation predictions. Team confirmation remains pending.

Is one threshold suitable for every emotion?

Planned controlled decoding comparison • no new test results claimed

Freeze the best neural checkpoint: current PhoBERT-27

FIXED THRESHOLD

Use $\tau = 0.5$ for every emotion.

GLOBAL TUNED

Select one shared threshold on validation.

PER-LABEL TUNED

Select a separate threshold for each emotion.

1. Tune on validation

Compare decoders; review FP / FN

→

2. Lock all choices

Checkpoint + thresholds + protocol

→

3. Evaluate test once

Report scores and limitations

TAKEAWAY **Hold model weights fixed; change the decoder, then evaluate held-out test once.**

Source: course Final RQ1 and current midterm protocol. Thresholds selected strictly on validation predictions.

What the current study supports

Three evidence-backed observations; a focused next experiment

01 PhoBERT leads within both label spaces

Macro-F1: 0.5974 over 27 emotions; 0.5783 over all 28 labels.

02 Neutral is not a shared-emotion F1 gain here

The 28-output projection is 1.38 pp lower than PhoBERT-27.

03 Evaluate decoding with a frozen backbone

Explore fixed, global and per-label thresholds without retraining.

SCOPE AND LIMITATIONS

One seed; validation-based selection.

No model test scores or significance claims.

Original stratification algorithm unverified.

Heuristic styles; error review needs confirmation.

TAKEAWAY **PhoBERT is the strongest model in each setting; Neutral and decoder design remain open questions.**

Sources: current Kaggle 08–09 artifacts; ViGoEmotions (EACL 2026, doi:10.18653/v1/2026.eacl-long.129); PhoBERT (2020).